

4.5.3 Units of information

4.5.3.1 Bits and bytes

Content	Additional information
Know that: - the bit is the fundamental unit of information - a byte is a group of 8 bits.	A bit is either 0 or 1.
Know that the 2^n different values can be represented with n bits.	For example, 3 bits can be configured in $2^3 = 8$ different ways. 000, 001, 010, 011, 100, 101, 110, 111

4.5.3.2 Units

Content	Additional information
Know that quantities of bytes can be described using binary prefixes representing powers of 2 or using decimal prefixes representing powers of 10, eg one kibibyte is written as $1\text{KiB} = 2^{10}\text{ B}$ and one kilobyte is written as $1\text{ kB} = 10^3\text{ B}$. Know the names, symbols and corresponding powers of 2 for the binary prefixes: • kibi, Ki - 2^{10} • mebi, Mi - 2^{20} • gibi, Gi - 2^{30} • tebi, Ti - 2^{40} Know the names, symbols and corresponding powers of 10 for the decimal prefixes: • kilo, k - 10^3 • mega, M - 10^6 • giga, G - 10^9 • tera, T - 10^{12}	Historically the terms kilobyte, megabyte, etc have often been used when kibibyte, mebibyte, etc are meant.